

Session 3

1) Creating a Virtual Machine

- * VMware environment / workstation Pro 12 was installed and launched.
- * A new virtual machine was created using the custom option.
- * Installation model was selected using an ISO file.
- * The guest operating system and its version were chosen.
- * The virtual machine was configured by allocating RAM, CPU core and disk space.
- * The set up was completed and operating system was installed successfully.

2) Installing Python on the Virtual Machine

- * The VM was started and the user logged into the operating system.
- * For Linux: Python and pip were installed using terminal commands.

For Windows: Python installer was downloaded.

3) Running a simple Python Program

- * A python script was created to implement a basic machine learning model inside the Virtual Machine.
- * Required libraries such as NumPy, Pandas, scikit-learn was imported for data handling and model building.
- * A dataset was loaded and preprocessed by handling missing values and selecting relevant features.
- * The dataset was divided into training and testing sets to evaluate the model performance.

- * A suitable machine learning algorithm such as :
 - Linear Regression
 - Logistic Regressionwas selected and trained using the training data.

- * The trained model was then tested using the test dataset to make the predictions.

- * The performance of the model was evaluated using appropriate metrics such as accuracy, mean squared error or precision.

* The program was executed successfully and the output confirmed that the Machine Learning model was correctly working within the virtual environment

Session 4

Installation of CloudSim Simulator

CloudSim is a framework for simulating cloud computing environments and evaluating different cloud strategies.

It is implemented in Java and requires Java and Eclipse IDE for execution.

⇒ Prerequisites:-

Before installing CloudSim, the system must have:

- 1) Java Development Kit (JDK) 8 or higher
- 2) Eclipse IDE or any Java-supporting IDE
- 3) CloudSim Source Code
- 4) Apache Commons Math Library

Step 1 :- Install Java Development Kit

CloudSim is written in Java, so JDK must be installed.

- Download JDK from the official Oracle website
- Install using installation wizard
- Set environment variable Path to JDK bin directory
- Verify installation using command :

`java -version`

Step 2 :- Install Eclipse IDE

CloudSim requires an IDE to manage & run Java projects.

- Download Eclipse IDE
- Install and launch Eclipse
- Create a new Java workspace.

Step 3 :- Download CloudSim

CloudSim is open source and can be downloaded from GitHub

- Download CloudSim ZIP file
- Extract it to a directory such as `C:\CloudSim`

Step 4 :- Identify Modules Folder

Inside extracted folder open modules directory which contains :

- cloudsims (core library)
- cloudsims - examples (example programs)
- cloudsims - network (network module)

Copy the cloudsims folder

Step 5 :- Create Java Project in Eclipse

- Open Eclipse
- File → New → Java Project
- Enter project name
- Click Finish